

Lecture 11: Random Variables

Definition and Types

A random variable is ...

Examples:

Two major types of random variables:

- 1.

- 2.

Discrete random variables: two examples

Let X represent

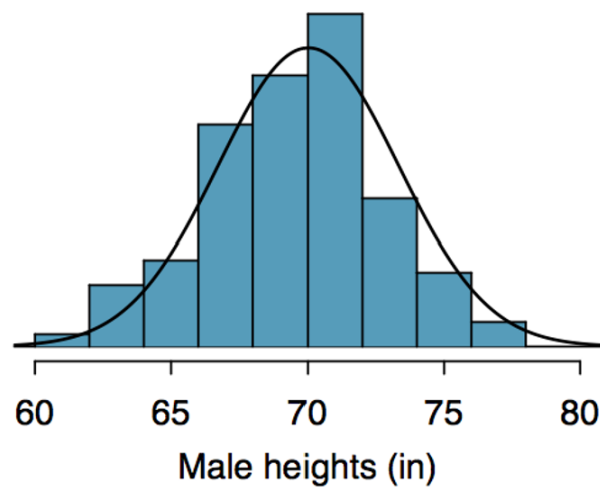
Value of X	1	2	3	4	5	6	7	8	9
Probability	.301	.176	.125	.097	.079	.067	.058	.051	.046

Let Y represent

Value of Y	1	2	3	4	5	6	7	8	9
Probability	1/9	1/9	1/9	1/9	1/9	1/9	1/9	1/9	1/9

Continuous random variables: one example

Let H represent

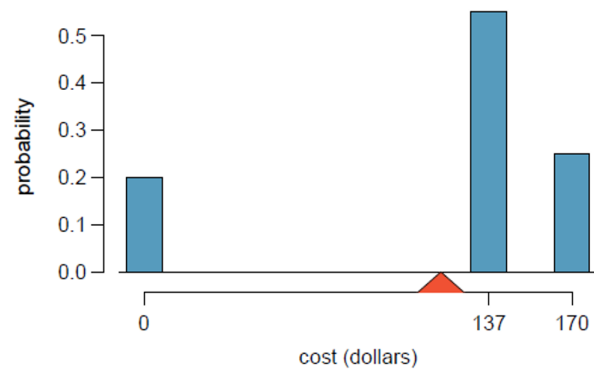


Mean and Variance for Discrete Random Variables

Example: bookstore revenue

i	1	2	3	Total
x_i	\$0	\$137	\$170	–
$P(X = x_i)$	0.20	0.55	0.25	1.00

Mean or expected value of a discrete random variable:



Random phenomenon: Bet \$1 on a game of Chuck-a-Luck (casino dice game) and let X represent your net gain or loss from that bet.

x_i	-1	1	2	5
$P(X = x_i)$	0.579	0.347	0.069	0.005

Expected gain/loss is:

Interpretation:

Random phenomenon: Take a SRS of size 3 from a large population that is $\frac{2}{3}$ female, $\frac{1}{3}$ male, and let X represent the number of females in the sample.

x_i	0	1	2	3
$P(X = x_i)$	1/27	6/27	12/27	8/27

Calculate $E[X]$.

Bookstore example revisited:

i	1	2	3	Total
x_i	\$0	\$137	\$170	–
$P(X = x_i)$	0.20	0.55	0.25	1.00

Variance of a discrete random variable:

Rules for mean and variance: example

x_i	1	2	3	4
$P(X = x_i)$	.1	.4	.4	.1

y_i	3	6	9	12
$P(Y = y_i)$	.1	.4	.4	.1

Rules for mean and variance: For random variables X and Y and constants a and b ,

Random phenomenon: Roll 2 “weird” 6-sided dice: one has faces with 1,3,4,5,6,8 pips; the other has faces with 1,2,2,3,3,4 pips. Let Z be the total number of pips on the two up-faces.

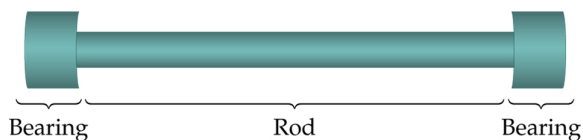
Random phenomenon: Toss a fair coin 4 times, and let X be the number of flips that come up heads and Y the number of flips that come up tails.

One more rule for variance:

Random phenomenon: My commute to work takes X minutes, where X is a random variable with mean 15 minutes and standard deviation 2 minutes. The same is true for my commute back home.

- Mean and variance of my total commute time in a day?
- Mean and variance of my total commute time in a workweek (5 days)?

Example: A mechanical assembly consists of a rod with a bearing on each end. The three parts are manufactured independently, and all vary a bit from part to part. Rods have mean length 12 cm and st. dev. 4 mm. Bearings have mean length 2 cm and st. dev. 4 mm.



What are the mean and st. dev. of the total length of the assembly?